Rhys Inward

UK Field Epidemiology Training Programme fellow, UK Health Security Agency.
DPhil candidate in Biology, University of Oxford.

rhys.inward@biology.ox.ac.uk rhysinward.github.io ORCID 0000-0003-0016-661X
github.com/rhysinward

Positions

Sep 2026 – present	**UK Field Epidemiology Training Programme fellow** UK Health Security Agency, North West Field Service, Liverpool Applied field epidemiology training, covering surveillance, outbreak response and applied projects.
Aug 2021 – 2025	**Data scientist in genomic epidemiology** Department of Biology (formerly Zoology), University of Oxford Genomic epidemiology of COVID-19 variants and control measures, scalable phylodynamic pipelines, dengue resurgence in Vietnam, and biases in COVID-19 surveillance data with the Global.health consortium.
May – Nov 2020	**Research assistant** Roslin Institute, University of Edinburgh Phylogenetic analysis of the spread of SARS-CoV-2 in Scotland.

Education

Oct 2023 – present	**DPhil in Biology** University of Oxford, Department of Biology and Pandemic Sciences Institute Thesis on integrating multiple data sources and quantitative tools to understand pathogen dynamics, with chapters on dengue and chikungunya transmission and on the GRAPEVNE pipeline environment. Supervised by Moritz Kraemer and Katrina Lythgoe. Funded by the Oxford-NaturalMotion Graduate Scholarship with Jesus College.
Sep 2021 – Aug 2022	**PGCert in Philosophy** University of Cambridge
Oct 2020 – Oct 2021	**MSc Epidemiology, Distinction** Imperial College London, School of Public Health Dissertation on genomic sampling strategies for estimating SARS-CoV-2 epidemiological parameters, supervised by Nuno Faria and Kris Parag and later published in Nature Communications. Gold Prize at the European stage of the EIT Innovation in Health challenge.

Sep 2017 – Jul 2020

BSc (Hons) Infectious Diseases, First Class

University of Edinburgh

Ritchie Prize for the most distinguished student in the year. My dissertation, on the spread and evolution of zoonotic RNA viruses from China, received the highest mark in the cohort.

Publications

Joint first authors are marked *. Full list on Google Scholar.

Peer-reviewed articles

2026 — Khurana MP, Tsui JLH, Gutierrez B, ..., **Inward RPD**, ..., Kraemer MUG. Global approaches to infectious disease surveillance and modeling. *Nature Medicine*. doi:10.1038/s41591-026-04351-4 (review)

2026 — Stewart CR, Lyon J, Nash PS, Best JG, Bonifacio G, Jitpimolmard J, **Inward RPD**, Chan E, Oliver R, Werring DJ. Cardiac structure relates to hemorrhagic cerebral small vessel disease phenotype. *Journal of the American Heart Association*. doi:10.1161/JAHA.124.039474

2026 — Tsui JLH, Sambaturu P, Pena RE, Too L, Gutierrez B, **Inward RPD**, Kraemer MUG, du Plessis L, Pybus OG. Transmission lineage dynamics and the detection of viral importation in emerging epidemics. *Epidemics*. doi:10.1016/j.epidem.2026.100893

2025 — Dasgupta A, Perez-Fernandez I, Huynh T, ..., **Inward RPD**, ..., Kraemer MUG. Scalable, open-access and multidisciplinary data integration pipeline for climate-sensitive diseases. *Wellcome Open Research*. doi:10.12688/wellcomeopenres.24774.3 (software)

2025 — Chabuka L, Choga WT, Mavian CN, ..., **Inward RPD**, ..., Kagoli M. Genomic surveillance of climate-amplified cholera outbreak, Malawi, 2022–2023. *Emerging Infectious Diseases*. doi:10.3201/eid3106.240930

2025 — Brittain JS*, Tsui J*, **Inward RPD***, ..., Kraemer MUG. GRAPEVNE – Graphical Analytical Pipeline Development Environment for Infectious Diseases. *Wellcome Open Research*. doi:10.12688/wellcomeopenres.23824.1 (joint first author, software)

2024 — Cadwgan J, Goodwin J, Babcock B, ..., **Inward RPD**, ..., UK Childhood Neurological Disorders PSP Group. UK research priority setting for childhood neurological conditions. *Developmental Medicine & Child Neurology*. doi:10.1111/dmcn.16021 (ppie)

2024 — Gutierrez B, Tsui JLH, Pullano G, ..., **Inward RPD**, ..., Kraemer MUG. Routes of importation and spatial dynamics of SARS-CoV-2 variants during localized interventions in Chile. *PNAS Nexus*. doi:10.1093/pnasnexus/pgae483

2024 — Tsui JLH, Pena RE, Moir M, **Inward RPD**, ..., Sambaturu P. Impacts of climate change-related human migration on infectious diseases. *Nature Climate Change*. doi:10.1038/s41558-024-02078-z

2023 — Tran VT*, **Inward RPD***, Gutierrez B, Nguyen NM, Nguyen PT, Rajendiran I, Cao TT, Duong KTH, Kraemer MUG, Yacoub S. Reemergence of Cosmopolitan genotype dengue virus serotype 2, southern Vietnam. *Emerging Infectious Diseases*. doi:10.3201/eid2910.230529 (joint first author)

2023	Castelán-Sánchez HG*, Delaye L*, **Inward RPD***, Dellicour S*, Gutierrez B*, ..., Escalera Zamudio M. Comparing the evolutionary dynamics of predominant SARS-CoV-2 virus lineages co-circulating in Mexico. *eLife*. doi:10.7554/eLife.82069 (joint first author)
2023	Tsui JLH*, McCrone JT*, Lambert B*, Bajaj S*, **Inward RPD***, Bosetti P*, ..., Kraemer MUG. Genomic assessment of invasion dynamics of SARS-CoV-2 Omicron BA.1. *Science*. doi:10.1126/science.adg6605 (joint first author)
2023	Escalera-Zamudio M, Kosakovsky Pond SL, Martínez de la Viña N, Gutiérrez B, **Inward RPD**, Thézé J, van Dorp L, Castelán-Sánchez HG, Bowden TA, Pybus OG, Hulswit RJG. Identification of evolutionary trajectories shared across human betacoronaviruses. *Genome Biology and Evolution*. doi:10.1093/gbe/evad076
2022	**Inward RPD***, Jackson F*, Dasgupta A, Lee G, Battle AL, Parag KV, Kraemer MUG; Global.health consortium. Impact of spatiotemporal heterogeneity in COVID-19 disease surveillance on epidemiological parameters and case growth rates. *Epidemics*. doi:10.1016/j.epidem.2022.100627 (joint first author)
2022	McCrone JT, Hill V, Bajaj S, ..., **Inward RPD**, ..., Kraemer MUG. Context-specific emergence and growth of the SARS-CoV-2 Delta variant. *Nature*. doi:10.1038/s41586-022-05200-3
2022	**Inward RPD**, Parag KV, Faria NR. Using multiple sampling strategies to estimate SARS-CoV-2 epidemiological parameters from genomic sequencing data. *Nature Communications*. doi:10.1038/s41467-022-32812-0 (first author)

Preprints

2025	Martínez-Duque P, Jiménez-Rico MA, Bacab-Cab LA, ..., **Inward RPD**, ..., Escalera-Zamudio M. A market-based virus monitoring strategy complementary to epidemiological surveillance in urban settings. *medRxiv*. doi:10.64898/2025.12.22.25342882
2025	de Abreu R, Fernandez IP, Mishra S, ..., **Inward RPD**, ..., Sparrow S. The role of climate change in the expansion of dengue. *medRxiv*. doi:10.1101/2025.10.06.25337235
2021	Lycett SJ, Hughes J, McHugh MP, ..., **Inward RPD**, ..., Robertson DL. Epidemic waves of COVID-19 in Scotland: a genomic perspective on the impact of the introduction and relaxation of lockdown on SARS-CoV-2. *medRxiv*. doi:10.1101/2021.01.08.20248677

Grants and awards

2026	**Bowers Fund grants (two awards), £300 each**
2025	**Bowers Fund grant, £300**
2025	**Jesus College Graduate Scholarship** Jesus College, Oxford
2023	**Prize for best speaker** EScamps conference, University of Cambridge
2023	**Oxford-NaturalMotion Graduate Scholarship** University of Oxford, in partnership with Jesus College
2023	**Graduate Choral Scholarship** St Peter's College, Oxford

2020–21	**Gold Prize, EIT Innovation in Health challenge** European Institute of Innovation and Technology. Winner at the local and European stages.
2020–21	**Runner-up, Imperial Healthcare Hackathon** Imperial College London
2020	**Ritchie Prize** University of Edinburgh. For the most distinguished student in the year, with the highest overall mark.
2020	**Highest dissertation mark** BSc Infectious Diseases, University of Edinburgh
2019	**Undergraduate Vacation Scholarship** Medical Research Scotland. Funded a summer research project on host–pathogen interactions.

Talks and posters

2026	**Climate, importation, and immunity shape chikungunya transmission across emerging and endemic settings** Poster, International Pandemic Sciences Conference, Singapore
2026	**PPIE: the patient experience** Guest speaker, NIHR Health and Care Internship Training Day, London
2025	**Championing patient and public involvement in research** Guest speaker, Encephalitis International VIP Changemaker Event, London
2025	**Incorporating patient voice in clinical research** Expert panellist, Evelina London Research Showcase, London
2025	**Global homogenisation of dengue virus diversity following the relaxation of COVID-19 pandemic restrictions** Poster, Ecology and Evolution of Infectious Diseases (EEID), University of Notre Dame, USA
2024	**Patient perspective: from patient to researcher** Invited speaker and expert panellist, British Paediatric Neurology Association Annual Meeting, Birmingham
2024	**Patient perspective: from patient to researcher** Invited speaker, Royal College of Paediatrics and Child Health, London
2023	**Streamlining large-scale analyses: building scalable and flexible pipelines for disease analytics – case study of dengue in Vietnam** Poster, Epidemics Conference, Bologna, Italy
2023	**Reemergence of Cosmopolitan genotype dengue virus serotype 2, southern Vietnam** Contributed talk, EScamps, University of Cambridge. Best speaker prize

2023 **Cryptic transmission and re-emergence of Cosmopolitan genotype of dengue virus serotype 2 within Ho Chi Minh City and southern Vietnam**
Poster, Virus Evolution and Molecular Epidemiology (VEME), Stellenbosch University, South Africa

2023 **Cryptic transmission and re-emergence of Cosmopolitan genotype of dengue virus serotype 2 within Ho Chi Minh City and southern Vietnam**
Poster, Pandemic Sciences Conference, University of Oxford

2022 **Viral evolution and genomic dataset construction; applying statistical inference while accounting for phylogenetic structure; Bayesian statistics for phylogenetics (BEAST)**
Invited lectures, Genomic Epidemiology and Statistics Workshop, Oxford University Clinical Research Unit, Ho Chi Minh City, Vietnam

2022 **Spatial dynamics of SARS-CoV-2 across a heterogeneous mobility landscape in Chile**
Poster, Virus Genomics and Evolution (VGE), Cambridge

2022 **Virus evolution and genomic dataset construction**
Invited lecture, Genomic Epidemiology and Data Science Workshop, Universidad Autónoma de Yucatán, Mérida, Mexico

2022 **Using multiple sampling strategies to estimate SARS-CoV-2 epidemiological parameters from genomic sequencing data**
Contributed talk, Ecology and Evolution of Infectious Diseases (EEID), Emory University, Atlanta, USA

Teaching

Jan 2025 – Jan 2026 **Non-Stipendiary Lecturer in Biology**
St Catherine's College, Oxford
Undergraduate tutorials in infectious disease epidemiology and evolution. I designed the teaching materials, set essays and marked assessments.

Jan 2025 – 2026 **Undergraduate tutor**
Jesus College, Oxford
Small-group tutorials in epidemiology and molecular biology.

Jan 2025 – 2026 **Module co-lead**
Department of Biology, University of Oxford
Co-developed the computational epidemiology practicals, written in R, for second-year MBiol students.

2022 **Genomic epidemiology workshops**
OUCRU, Ho Chi Minh City, Universidad Autónoma de Yucatán, Mérida
Invited lectures on virus evolution and genomic dataset construction, statistical inference that accounts for phylogenetic structure, and Bayesian phylogenetics with BEAST.

Jan 2022 – 2026

Demonstrator
Department of Biology (formerly Zoology), University of Oxford
Undergraduate practicals in phylogenetics, epidemiology and evolutionary biology, including help with coding in R.

Patient and public involvement

2026 – present

Co-design lead, Q-CDSS
NIHR Invention for Innovation (i4i) FAST, UCL AI4coma lab
Leading stakeholder co-design for a hybrid classical–quantum decision-support tool to predict critical events such as cardiac arrest in intensive care patients. Funded by NIHR i4i FAST in 2026, with Applied Quantum Computing and Guy's and St Thomas' NHS Foundation Trust.

Aug 2024 – present

Consultant in patient and public involvement and engagement (PPIE)
Evelina London Children's Hospital, King's College London
Evaluating patient and public involvement in the Neurology department through surveys of clinical staff, and recommending how to build patient perspectives into research strategy and reporting.

2021 – present

Public contributor
NIHR Health Protection Research Unit in Emerging and Zoonotic Infections
Helping the unit improve how it involves patients and the public in its research.

2020 – 2023

Young adult representative
James Lind Alliance, Childhood Neurological Conditions Priority Setting Partnership
On the steering group of the partnership that identified the UK's top ten research priorities for childhood neurological conditions, and in the team that analysed the survey data. Published in Developmental Medicine & Child Neurology in 2024.

2018 – present

Support, information and research volunteer
Encephalitis International
Peer support, public engagement and patient-facing resources, including the charity's information page on CLIPPERS.

Leadership and service

2026 – present

MCR Sports Representative
Jesus College, Oxford

2025

MCR President
Jesus College, Oxford
Served three terms leading the college's graduate common room, and co-chaired the Conference of Common Rooms. Before that I was the MCR's Social Secretary (2023–2025).